JULIETA D. RUBIS
AI Engineer (LLMs • RAG • Real-Time ML • AWS)
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PROFESSIONAL SUMMARY

AI Engineer specializing in LLM-powered systems, Retrieval-Augmented Generation (RAG), and real-time machine learning pipelines. Experienced in building production-ready AI solutions on AWS, including fine-tuning large language models (Mistral, Claude) and deploying scalable applications using LangChain and vector databases. Strong background in distributed systems and data-driven AI, delivering measurable improvements in performance, efficiency, and decision-making.

CORE SKILLS

Languages: Python, SQL, JavaScript
Machine Learning: PyTorch, TensorFlow, Scikit-learn, LSTM, Autoencoders
LLMs & GenAI: LangChain, Hugging Face, OpenAI API, AWS Bedrock, FAISS, RAG
Data Engineering: Apache Spark, Kafka, ETL Pipelines, Pandas, NumPy
Cloud & MLOps: AWS (SageMaker, EC2, S3, Bedrock), Docker, Git
Frameworks: Flask, Dash, Streamlit

PROFESSIONAL EXPERIENCE

Co-Founder & AI Engineer (CTO)
Techbottronix – Ireland
2023 – Present

- Co-founded an AI startup delivering real-time intelligent systems and predictive analytics solutions
- Designed and deployed AI-powered platforms for anomaly detection and operational optimization
- Built predictive maintenance systems using LSTM and Autoencoder models, improving early fault detection
- Developed real-time data pipelines using Apache Kafka and Spark for large-scale sensor data processing
- Led end-to-end system architecture across AI, backend, and cloud infrastructure

Key Achievements:

- Developed as part of MSc dissertation and extended into real-world application at Techbottronix
- Achieved 0.91 AUC and 87% recall in anomaly detection for wind turbine systems
- Processed 150,000+ real-time records, reducing false positives by 30%
- Delivered AI-driven solutions that improved operational decision-making and reduced downtime
- Presented at Google Developer Groups (GDG Portlaoise 2025)

Machine Learning Engineer (Intern)
Orcawise – Dublin, Ireland
Jun 2024 – Dec 2024

- Developed and deployed an LLM-powered EU AI Act compliance chatbot using LangChain, FAISS, and AWS Bedrock, enabling semantic search across 1,000+ documents and reducing information retrieval time by 60%
- Fine-tuned LLMs (Mistral-7B, Claude) using Hugging Face and AWS Bedrock, improving response relevance and contextual accuracy by 35% for domain-specific legal queries

- Contributed to a healthcare bias detection and analysis system, supporting data preprocessing, model evaluation, and visualization of fairness metrics

Key Achievements:
- Reduced information retrieval time by 60%
- Improved fairness evaluation efficiency by 40%

WordPress Developer (Intern)
Bord Na Móna – Ireland
Jul 2024 – Sep 2024
- Improved website performance, reducing load times by 30%
- Enhanced system stability by 40%
- Delivered and deployed 5+ production websites

EDUCATION

MSc Artificial Intelligence & Machine Learning
University of Limerick, Ireland | 2023 – 2025
Dissertation: AI-Powered Anomaly Detection and Predictive Maintenance

Higher Diploma – Artificial Intelligence Applications
College of Computing Technology | 2022 – 2023

Certificate in Software Engineering
Technological University of the Shannon | 2022 – 2022

Higher Diploma in Digital Marketing
Dublin Business School | 2021 – 2022

QQI Level 5 in Healthcare Support
B&B Nursing, Ireland | 2020 – 2021

BSc Information Technology
Jose Rizal University | 2005 – 2009
Dissertation: Payroll System using PHP and MySQL

CERTIFICATIONS
- AWS AI & ML Scholarship Nanodegree – Udacity (2025)
- AWS AI Programming with Python Scholarship Nanodegree Program – Udacity (2024)
- Python for Data Science and AI – IBM (2024)

ADDITIONAL EXPERIENCE
Healthcare Assistant - Elmhurst Nursing Home – Ireland | 2020 – 2022
- Maintained high-accuracy digital records in clinical systems
- Delivered patient care in high-responsibility environments

REFERENCES AVAILABLE ON REQUEST